

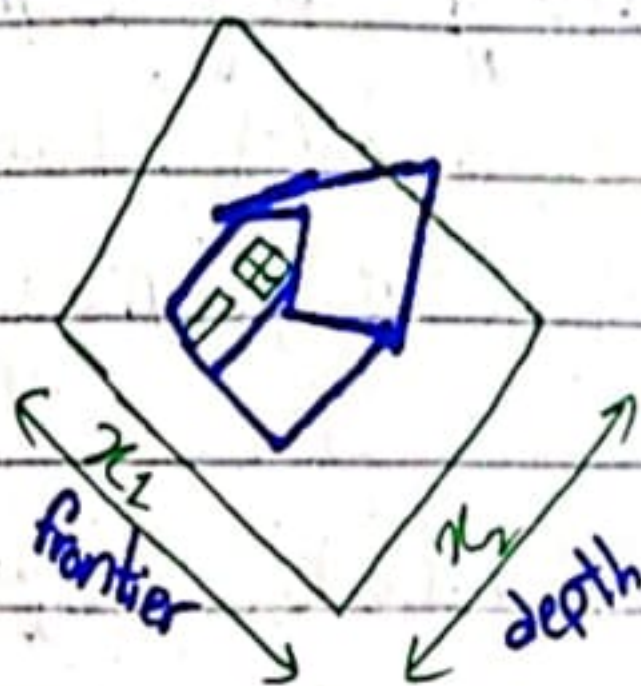
Feature Engineering

Choosing right features is a critical step to making algorithm work well.

example taken (predicting price of house).

- x_1 is width of plot size that the house is built on

↳ Frontier



- x_2 is depth of plot size

Given these two features you might build a model:

$$f(\vec{x}) = \overbrace{w_1 x_1}^{\text{Frontage}} + \overbrace{w_2 x_2}^{\text{depth}} + b$$

→ this model is okay but there's another way to how you choose a diff. way to use these features in a model that could be even more effective.

area = frontage x depth

↳ more predictive of price, than f & d as separate features.

So you may define a new feature:

$$x_3 = x_1 x_2$$

You can then have a model:

$$f_{\vec{w},b}(\vec{x}) = w_1x_1 + w_2x_2 + w_3x_3 + b$$

what we just did was,

Feature Engineering:

"Using intuition to design new features, by transforming or combining original features"

So instead of just using features you started with, engineering new features can result in much details.
